Finger Vein PAD Using Fractal Dimension and KNN

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Contents

- Background: Fractal dimension
- Methodology of three approaches
  - Approach 0 : Whole image FD
  - Approach 1 : Segment-wise FD
  - Approach 2 : FD around edges
- Results
  - Training, reducing training set
  - Substitution through synthetic data
- Conclusion

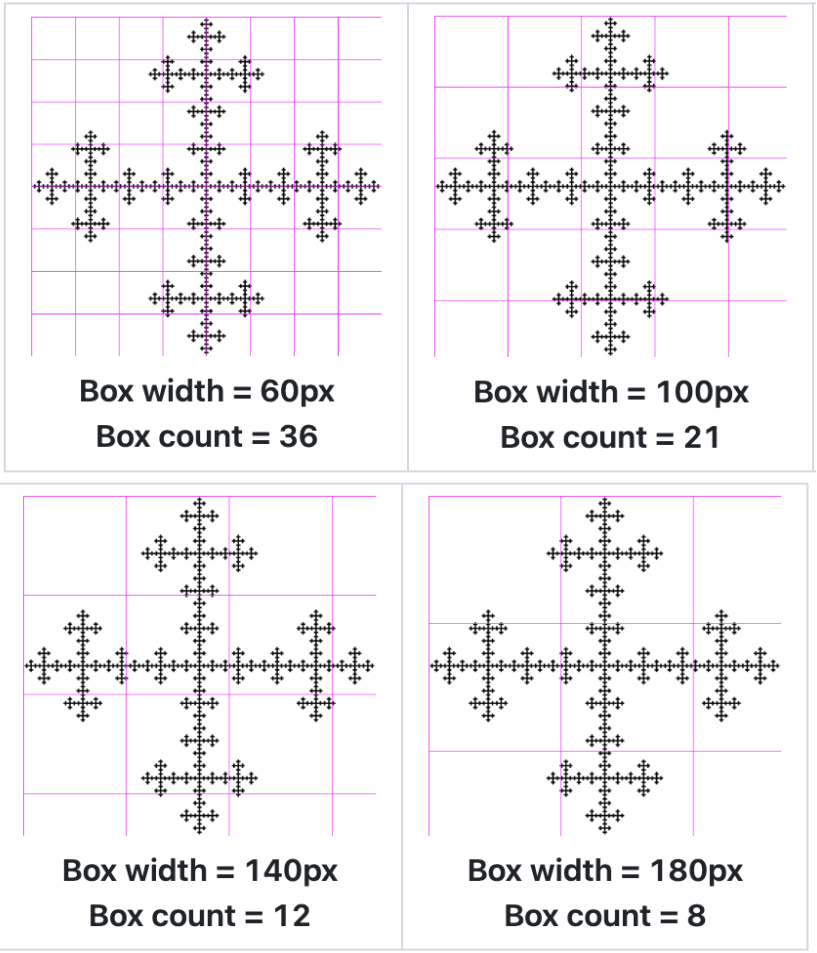
What is Fractal Dimension?

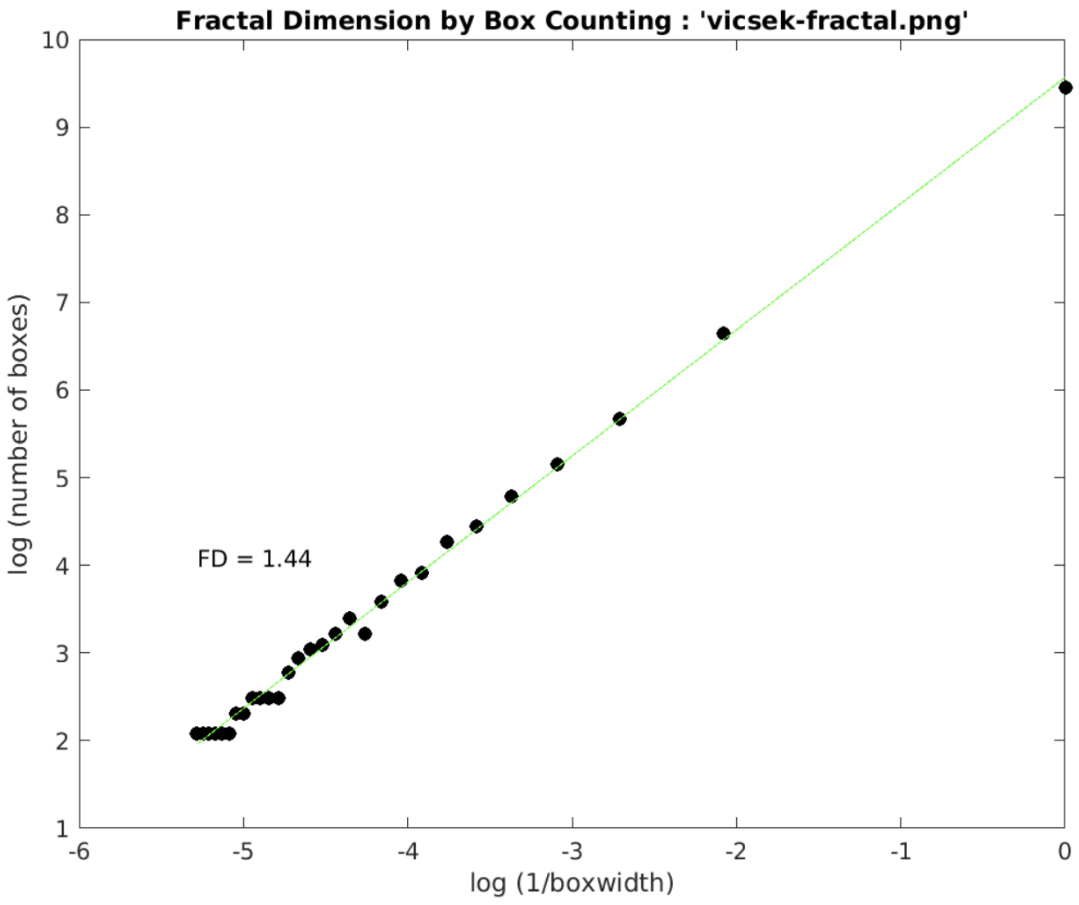
- **Simple definition**
  - Fractal Dimension (FD) is just a number that tells us how rough, messy or detailed an image is.
- **Not so simple definition**
  - Fractal Dimension measures 'complexity score' by tracking how the density of a shape increases as you zoom in, capturing the unique relationship between a fractal's detail and the scale at which it is observed.
- Technique used in our approach – **Box Counting method**
- **Low FD** → smooth, simple image
- **High FD** → rough, detailed, complex image
- FD is great for:
  - Texture analysis
  - Medical images
  - Face anti-spoofing
  - Natural patterns (skin, fingerprints, leaves)

Box counting Technique

The *box counting* method is a technique for approximating the fractal dimension of an object

- It can be viewed as zooming in or out of the image, to observe how detail changes with scale.

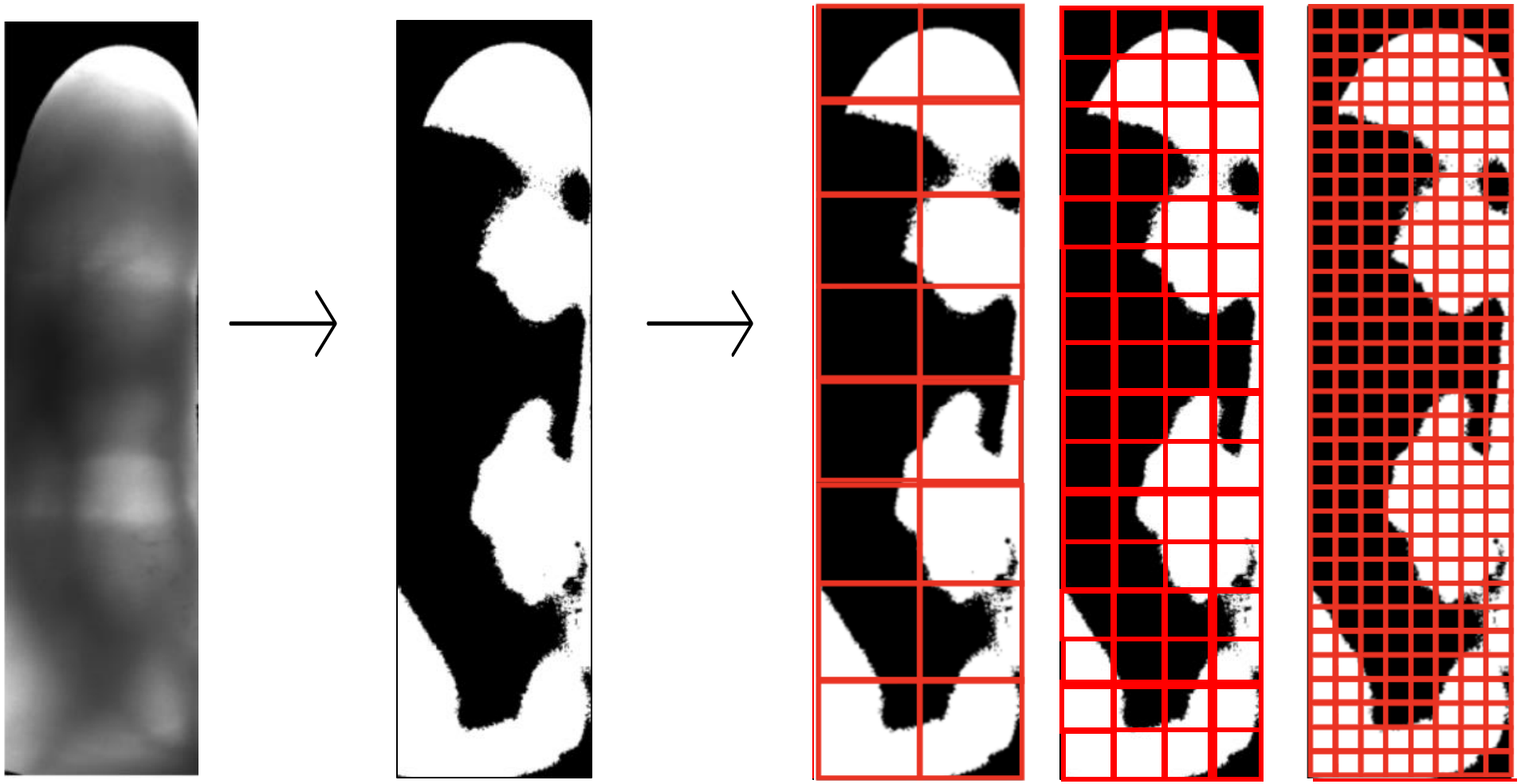




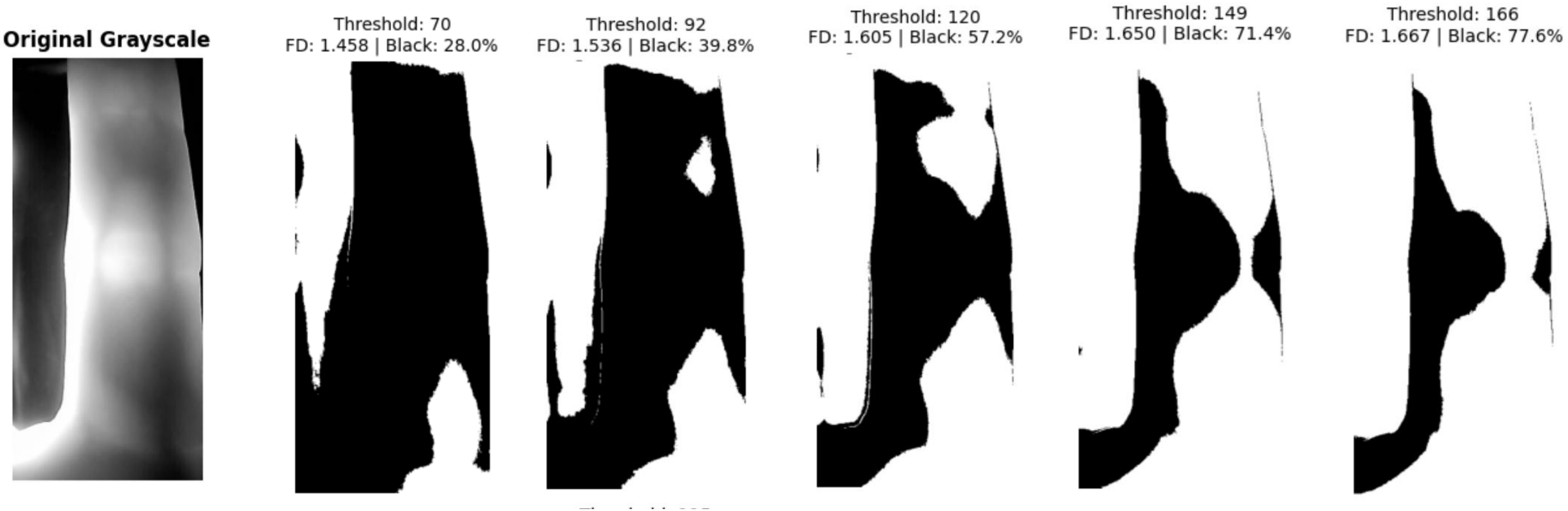
$$D = \lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}$$

ε : Size of box
N(ε) : No. of boxes to cover pattern.

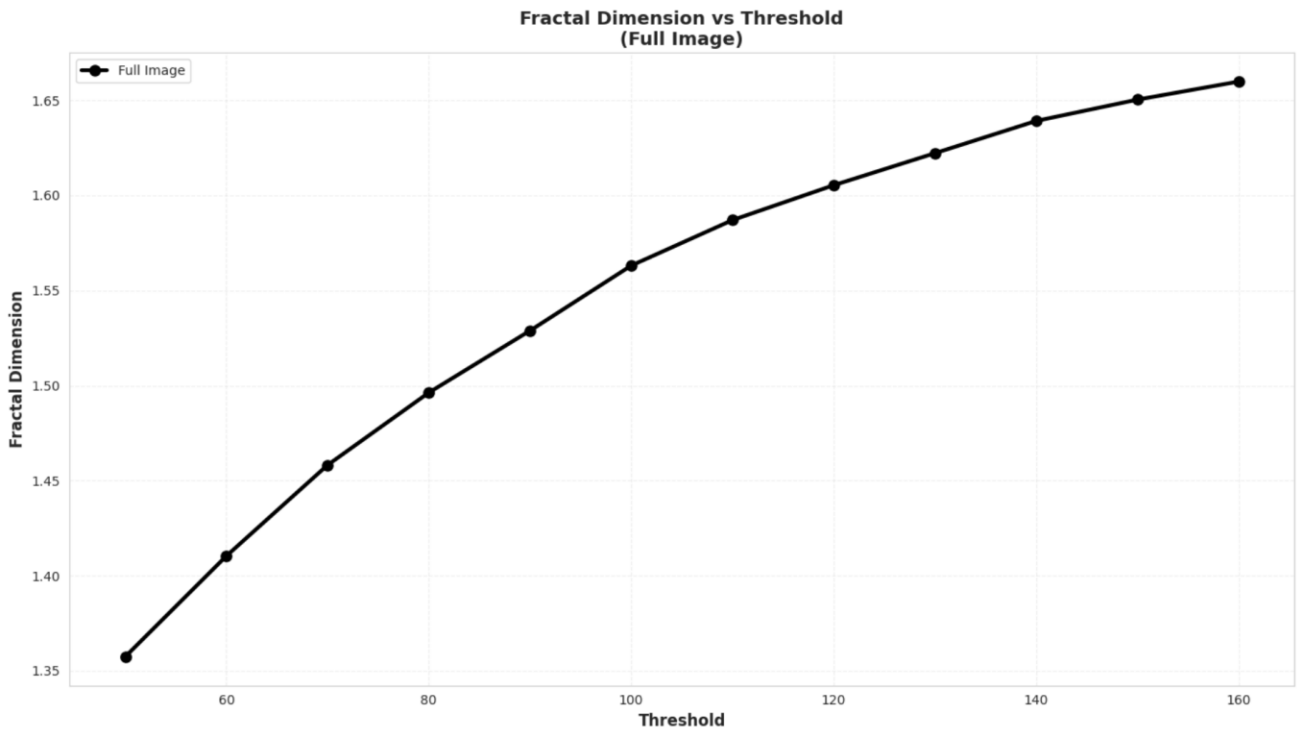
Box counting method for finger vein images



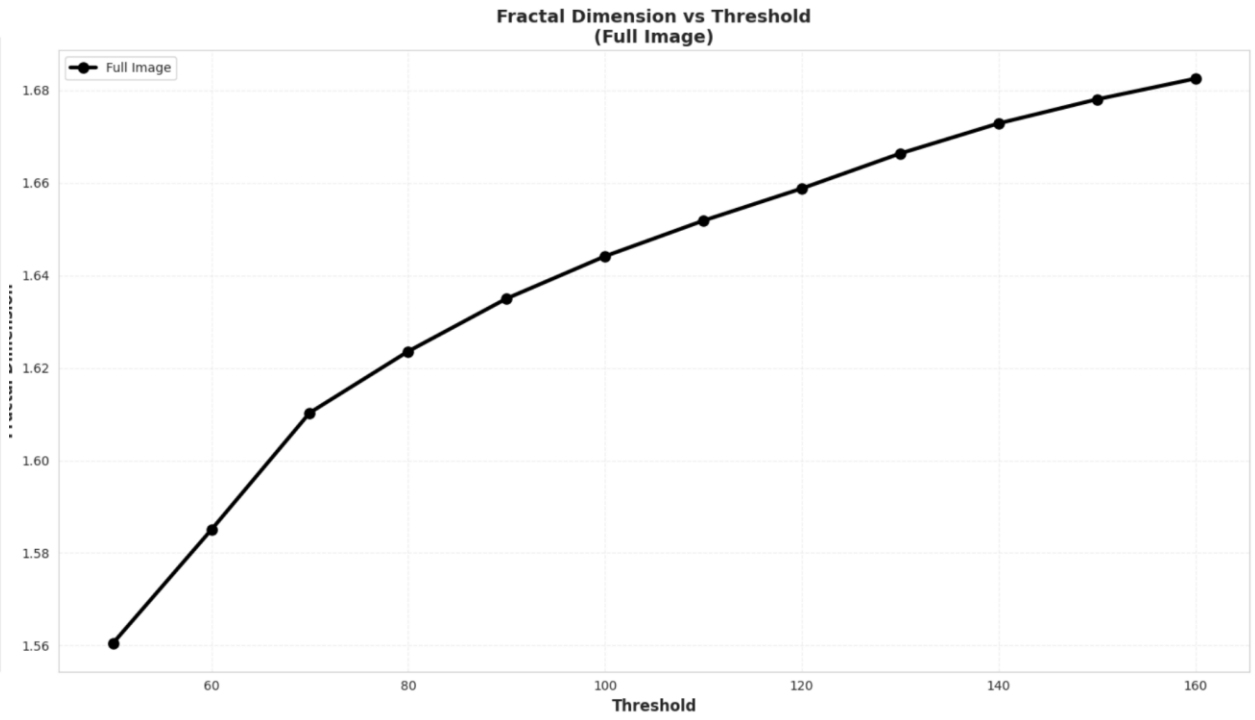
Approach 0 : FD of whole image



FD difference between real and spoof images

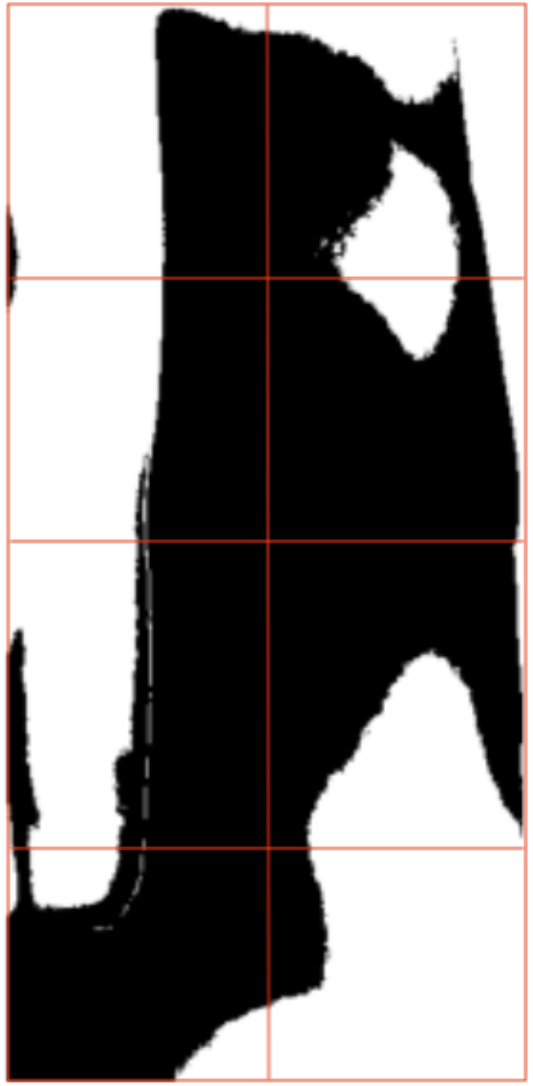


Real image



Spoof image

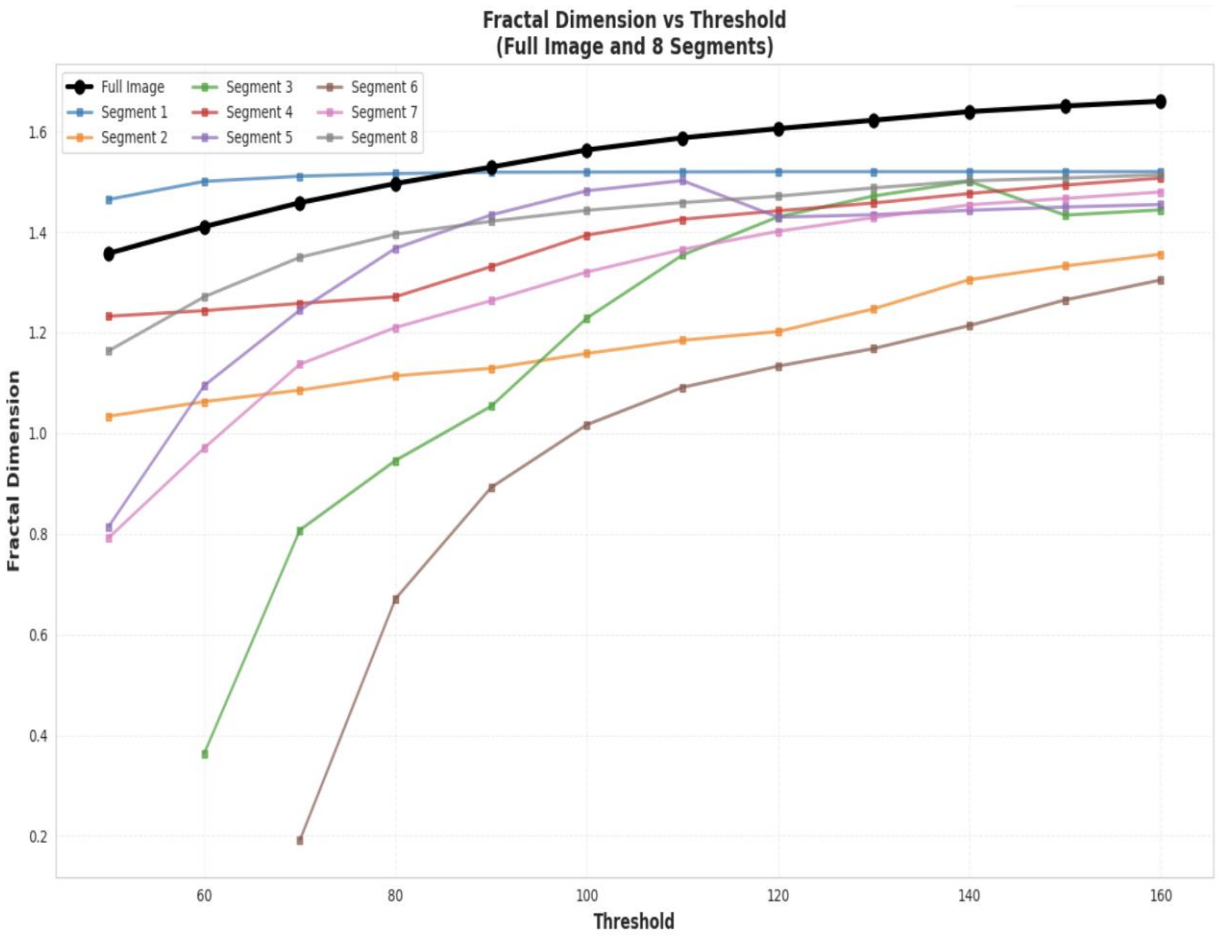
Approach 1 : FDs of segmented image



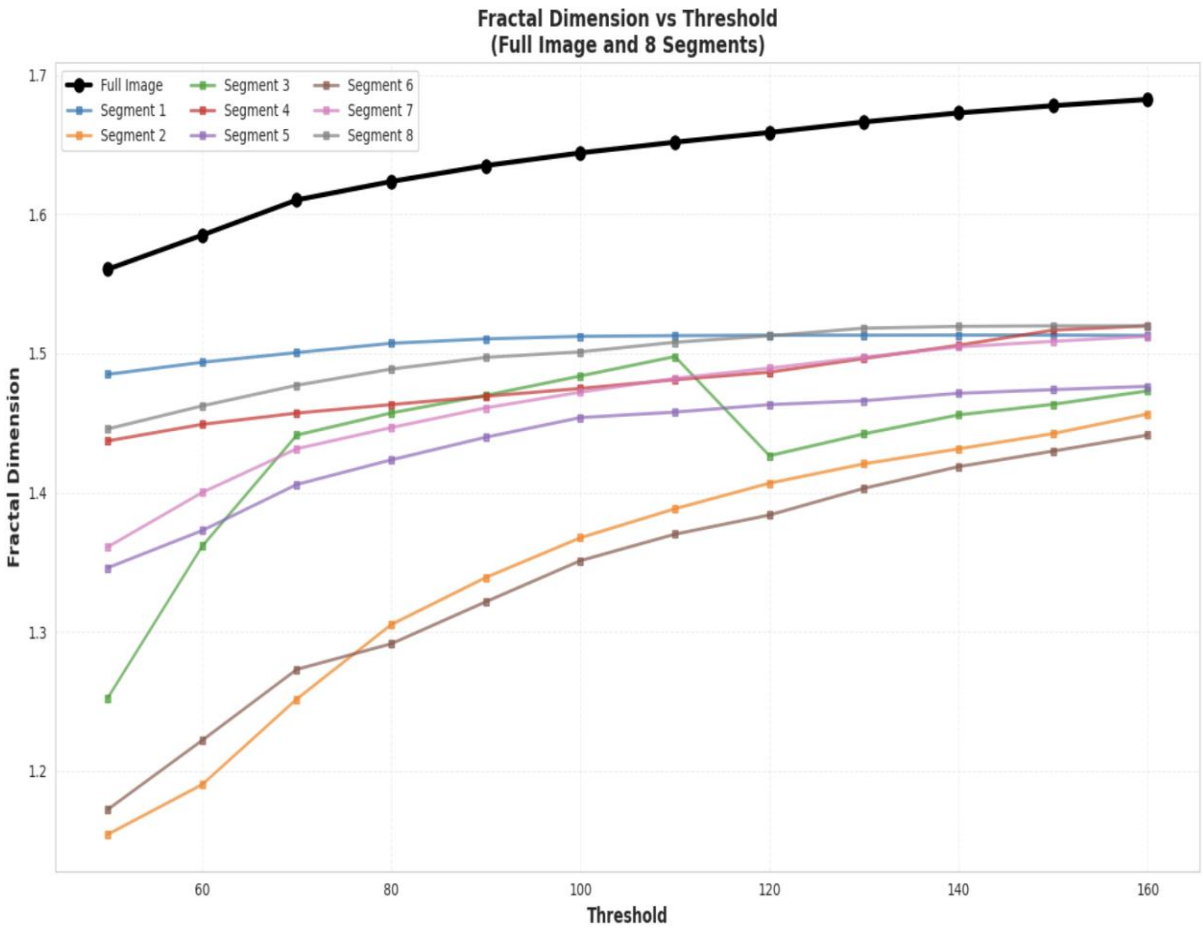
- The image is **segmented into 8 regions**.
- **Fractal Dimension (FD)** is computed for each segment at multiple **black–white (BW) thresholds**.
- Threshold values range from **50 to 130**.
- A total of **81 FD features per image** are extracted:
  - **9 FD values** from the full image (from approach 0)
  - **72 FD values** from segmented regions (9 thresholds × 8 segments)
- These features are used as input to the **KNN model**.

	A	B	C	D	E	F	G	H	I	J	K
1	fd_full_50	fd_seg1_50	fd_seg2_50	fd_seg3_50	fd_seg4_50	fd_seg5_50	fd_seg6_50	fd_seg7_50	fd_seg8_50	fd_full_60	fd_seg1_60
2	1.489872734	0.5344922959	1.316365983	1.440475336	1.718842314	1.14283682	1.253725307	1.429449416	1.718842314	1.532605581	0.8122809995
3	1.462268385	0.5033834823	0.6913822395	1.317360509	1.718842314	1.15856569	1.252950496	1.425602212	1.718842314	1.502582075	0.7346673524
4	1.463287807	0.4258861608	0.6267221378	1.274422592	1.718421703	1.245491549	1.222050465	1.443390695	1.718842314	1.507790256	0.613462581
5	1.462467138	0.5573969124	0.6189334945	1.254283841	1.718281244	1.264564419	1.203951596	1.435321834	1.718842314	1.508180274	0.965878155
6	1.484223053	0.8892771357	0.8841079307	1.502301659	1.718842314	1.24932734	1.037314017	1.459512942	1.718842314	1.522638963	1.069143579
7	1.355593066	1.45563608	0.1071428571	0.9715760307	1.370769681	1.443606066	1.167001308	1.334521234	1.365012495	1.461841741	1.51186708
8	1.371583119	1.488113216	0.6412342802	1.001995679	1.332134327	1.449736051	1.120483277	1.332771929	1.421296358	1.470521952	1.522924615
9	1.37754511	1.463990965	0.5783786777	1.017466347	1.326667836	1.44001118	1.098173881	1.369259317	1.466223179	1.468962859	1.49390977
10	1.403526521	1.481028089	0.2857142857	1.062152596	1.326323271	1.449513604	1.088552809	1.481027765	1.504185656	1.481245767	1.519483994
11	1.379704354	1.455752422	0.4785265604	1.020384704	1.351901484	1.435931266	1.152877914	1.353915614	1.476005307	1.474966421	1.501594044

FD differences for real & spoof segments



Real image



Spoof image

KNN training and testing steps

- Training and test data for each data set

DATA SET	Total files	Test files
PLUS	2193	439
SCUT	1440	288
IDIAP	240	40

- Five-fold cross validation
  - Step 0 : 4 Fold training, 1 Fold testing(real + spoof)
  - Step 1 : 2 Fold training, 1 Fold testing(real and spoof)
  - Step 2 : 1 Fold training, 1 Fold testing(real + spoof)
  - Step 3 : 1 Fold (real + spoof), 1 Fold (real + synthetic) training, 1 Fold testing
  - Step 4 : 1 Fold (real + spoof), 3 Fold (real + synthetic) training, 1 Fold testing
  - Step 5 : 4 Fold (real + synthetic) training, 1 Fold testing

Results – Segment wise FD - PLUS

Training set : Original	Accuracy	ACER	APCER	BPCER
Step 0 (4/5 training)	0.997	0.003	0.002	0.003
Step 1 (2/5 training)	0.997	0.002	0.000	0.004
Step 2 (1/5 training)	0.992	0.019	0.037	0.001

Training set : VAE + Original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.995	0.004	0.002	0.005
Step 4 (3/5 synthetic + 1/5 orig)	0.997	0.003	0.002	0.003
Step 5 (4/5 synthetic)	0.997	0.003	0.002	0.003

Training set : AE + Original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.995	0.006	0.007	0.005
Step 4 (3/5 synthetic + 1/5 orig)	0.994	0.013	0.023	0.003
Step 5 (4/5 synthetic)	0.993	0.014	0.026	0.003

Results – Segment wise FD - SCUT

Training set : Original	Accuracy	ACER	APCER	BPCER
Step 0 (4/5 training)	0.988	0.013	0.000	0.025
Step 1 (2/5 training)	0.973	0.027	0.000	0.054
Step 2 (1/5 training)	0.969	0.031	0.000	0.063

Training set : VAE + Original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.953	0.047	0.001	0.093
Step 4 (3/5 synthetic + 1/5 orig)	0.956	0.044	0.000	0.087
Step 5 (4/5 synthetic)	0.966	0.034	0.000	0.068

Training set : AE + Original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.960	0.040	0.000	0.081
Step 4 (3/5 synthetic + 1/5 orig)	0.960	0.040	0.000	0.080
Step 5 (4/5 synthetic)	0.972	0.028	0.000	0.057

Results – Segment wise FD - IDIAP

Training set : Original	Accuracy	ACER	APCER	BPCER
Step 0 (4/5 training)	0.883	0.118	0.000	0.235
Step 1 (2/5 training)	0.854	0.148	0.008	0.289
Step 2 (1/5 training)	0.817	0.183	0.008	0.359

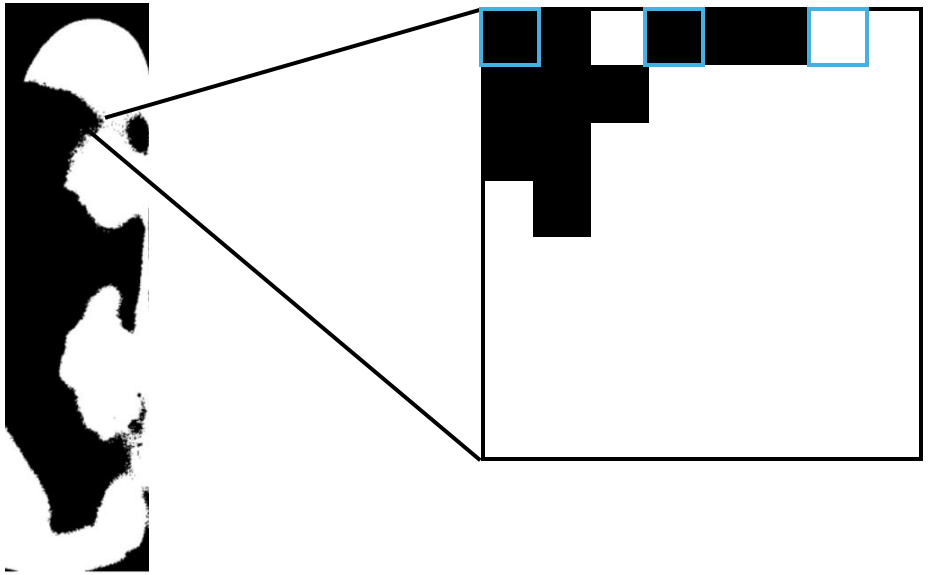
Training set : VAE + Original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.858	0.144	0.008	0.280
Step 4 (3/5 synthetic + 1/5 orig)	0.879	0.124	0.000	0.248
Step 5 (4/5 synthetic)	0.888	0.115	0.000	0.231

Training set : AE + Original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.846	0.157	0.017	0.297
Step 4 (3/5 synthetic + 1/5 orig)	0.871	0.133	0.000	0.265
Step 5 (4/5 synthetic)	0.879	0.123	0.000	0.247

Findings – FD with image segmentation

- The **PLUS (~99%)** and **SCUT (~96%)** datasets show superior performance compared to **IDIAP(~87%)**.
- Increasing the number of folds leads to consistent improvements in **accuracy and ACER** across all datasets.
- Performance remains comparable when using **VAE-generated and AE-generated synthetic data**.
- Replacing original data with synthetic data results shows **no substantial change in accuracy (±2%).**
- Could synthetic data substitute original data?

Approach 2 : Fractal dimension around edges

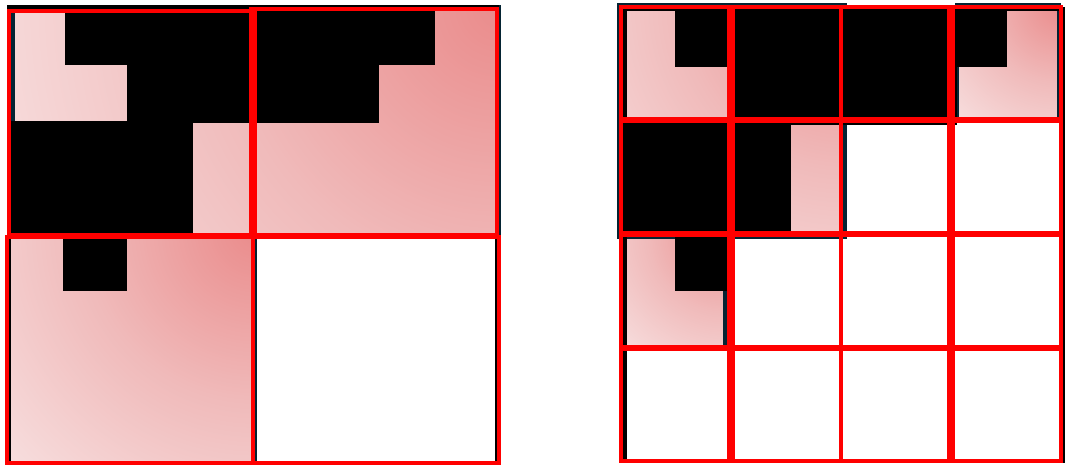


Stride: Window offset (3)

Binary image

Binary image
Closer look at
border region

Inside sliding window (15x15)

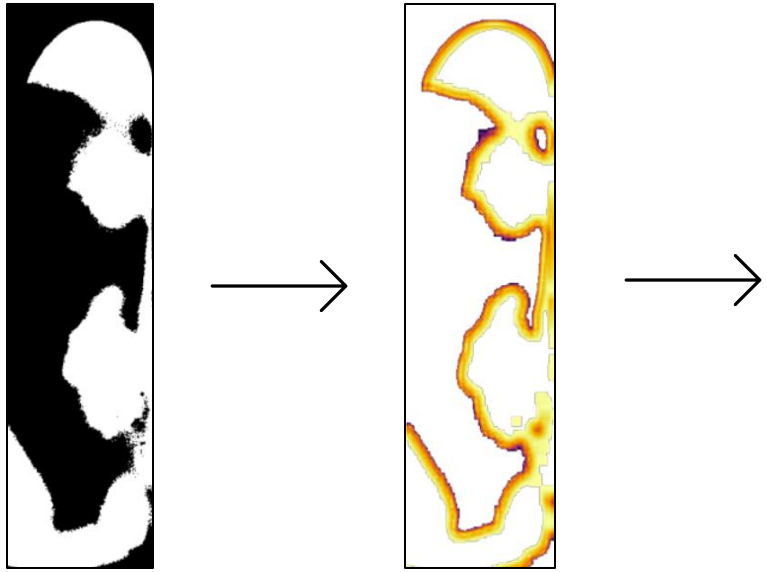


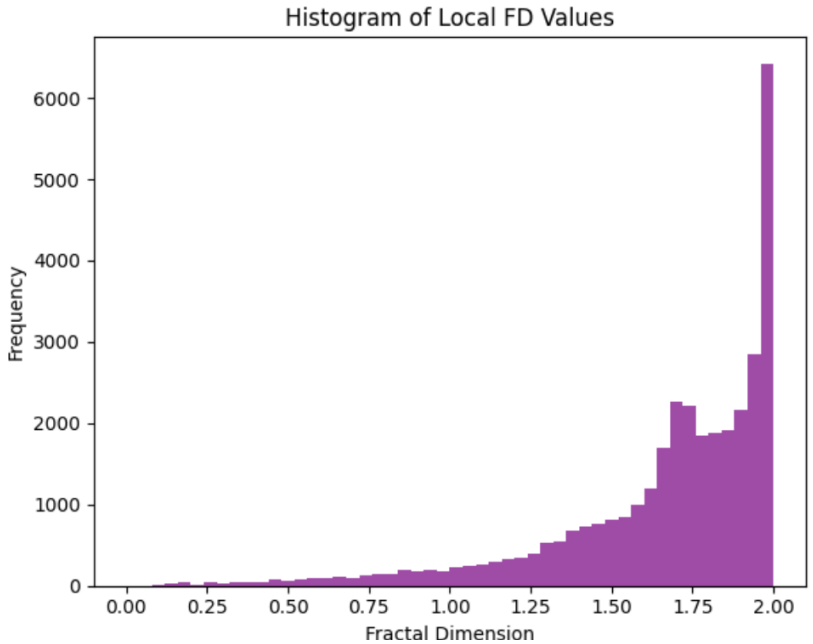
Boxes

Edge map
Pixels defining the border
Counting boxes with
border pixels

Edge map
Increasing box size
Scale parameter: 2, 4, 8

FD – histogram as feature vector

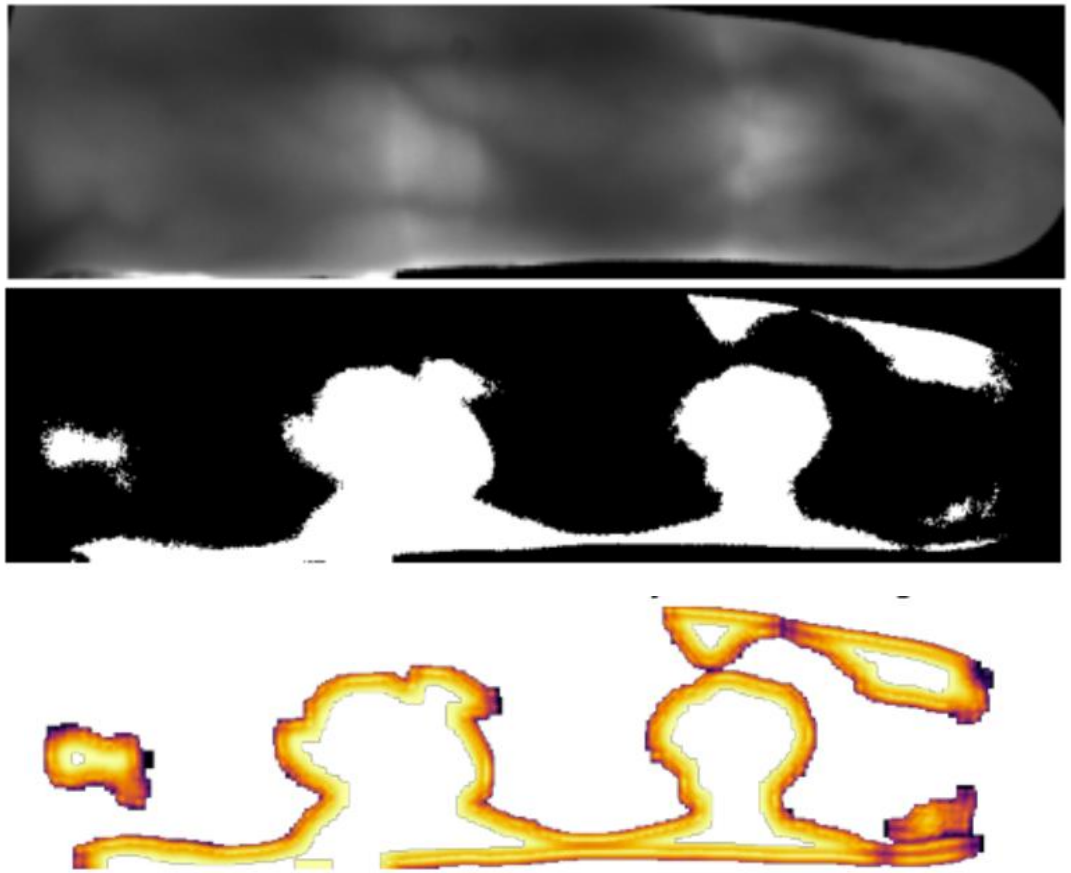


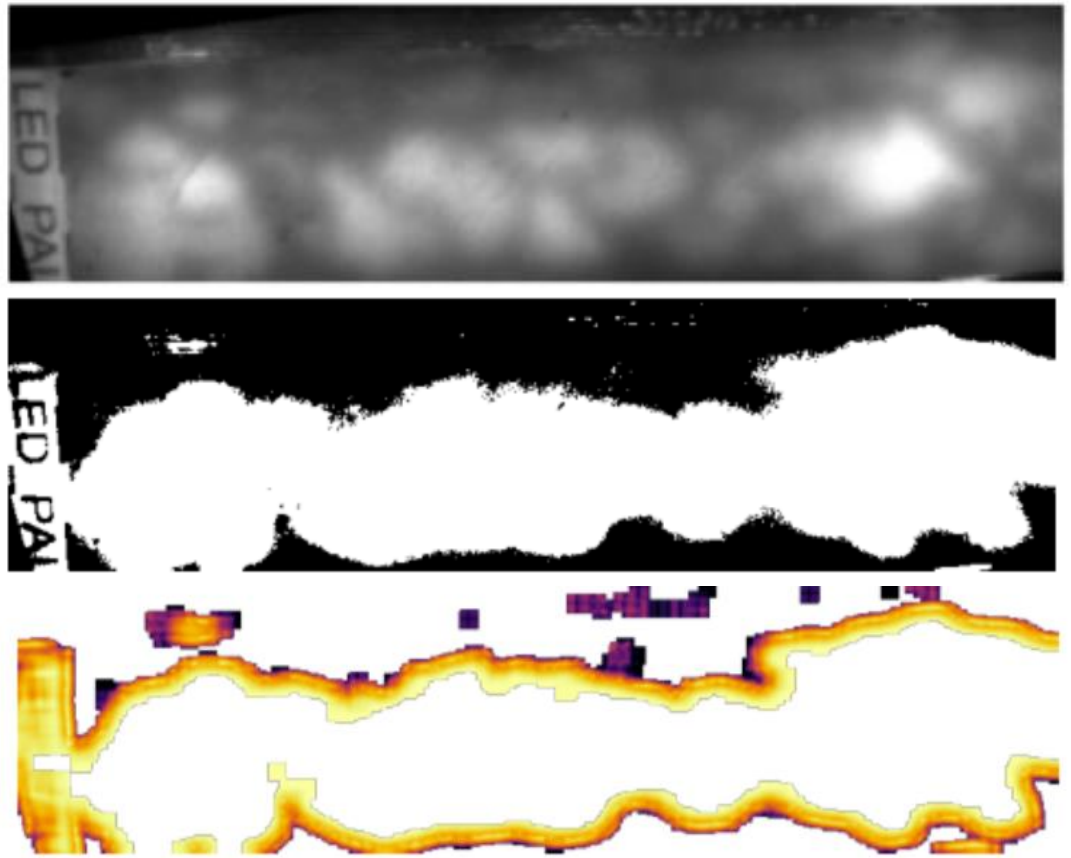


O	P	Q
bin_11	bin_12	bin_13
0.09073661635	0.004124391652	0.47430504
0.1239157373	0.006884207628	0.3958419386
0.06111894687	0	0.4983544899
0.1690102758	0.002253470344	0.4281593654
0.07507507508	0.01185395922	0.4781096886
0.1480923695	0.02761044177	0.4869477912
0.09719664416	0	0.4757519951
0.173214162	0.0173214162	0.4780710871
0.1153765471	0	0.5087056849
0.1603750308	0.008224360556	0.4646763714
0.1759324419	0.01172882946	0.5219329111
0.1579279848	0.007019021548	0.5474836808
0.1616883669	0	0.510594843
0.1391650099	0.004417936824	0.4108681246
0.2389858687	0.01558603491	0.5662926018
0.1317855541	0	0.3841408703
0.1957330202	0	0.5137991779

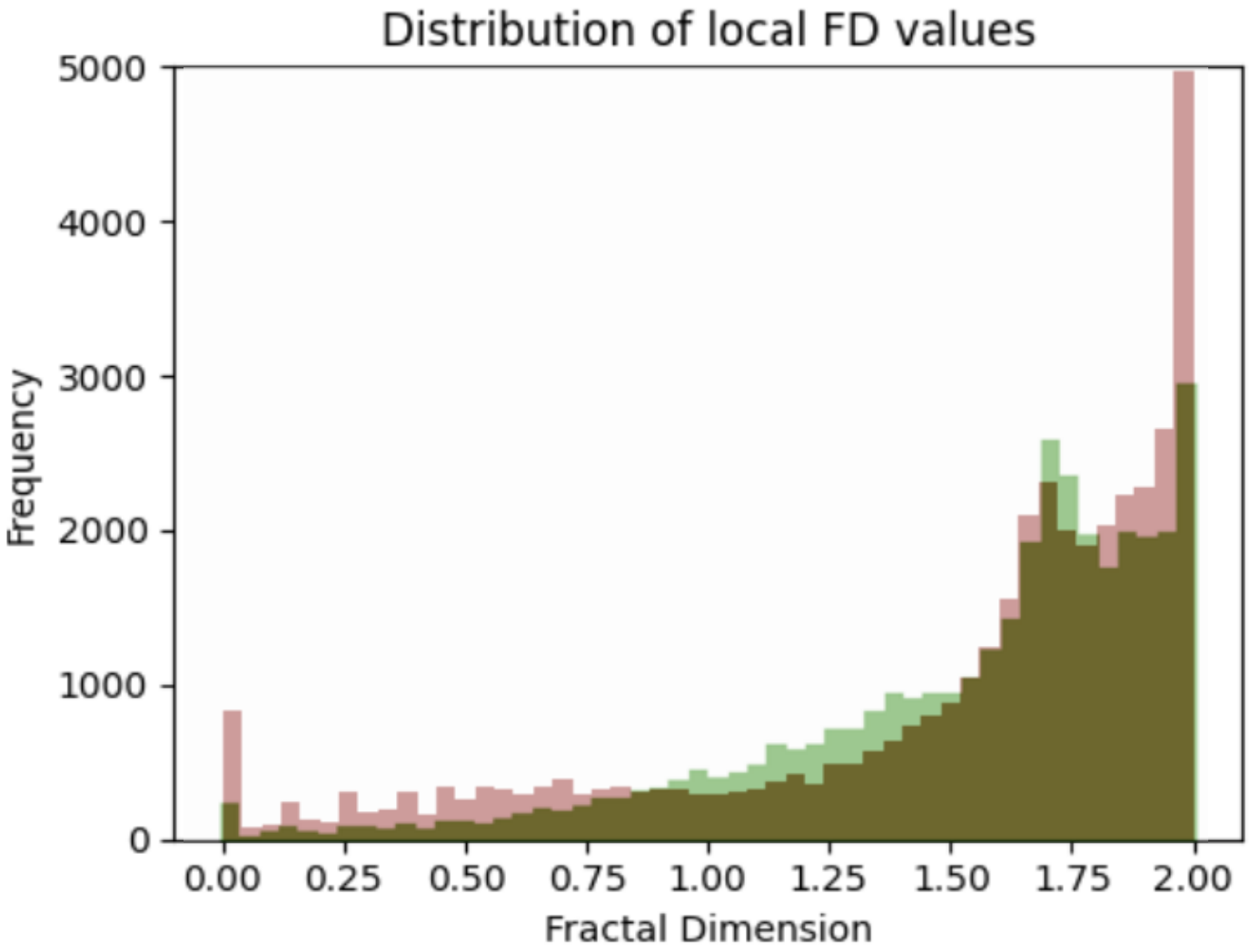
- Every window yields an FD value
- Using histogram to represent the distribution of FD values among edge regions
- Specifying bins: Dividing the FD value range into intervals
- Histogram shows relative frequency of occurring FD regions

Real vs spoof – PLUS dataset

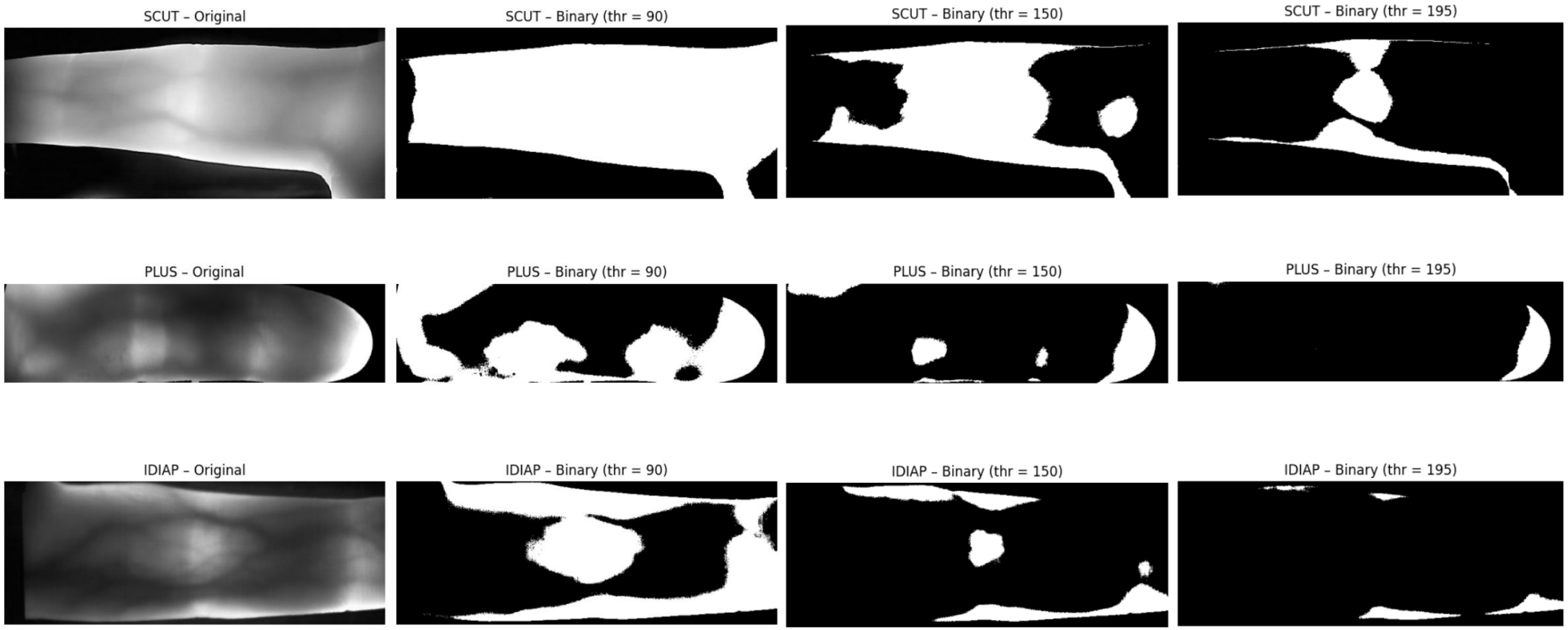




Real vs Spoof: PLUS dataset

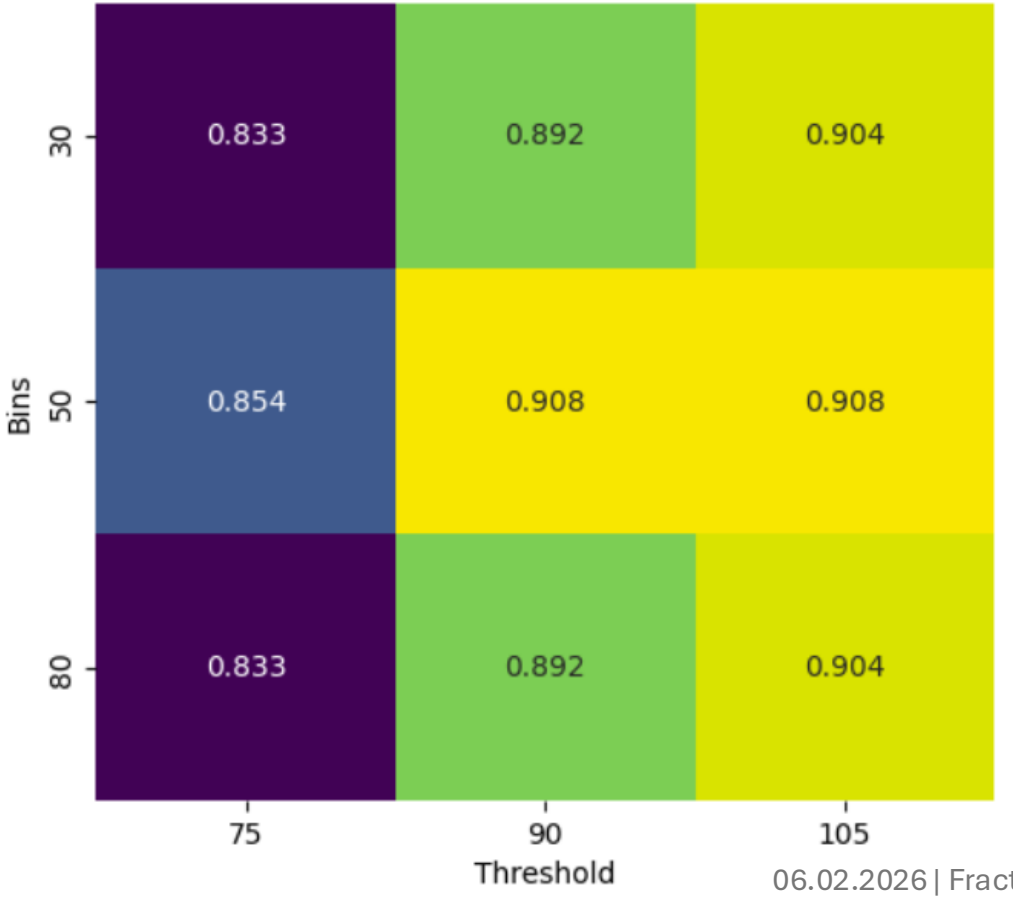


Choosing a binary threshold

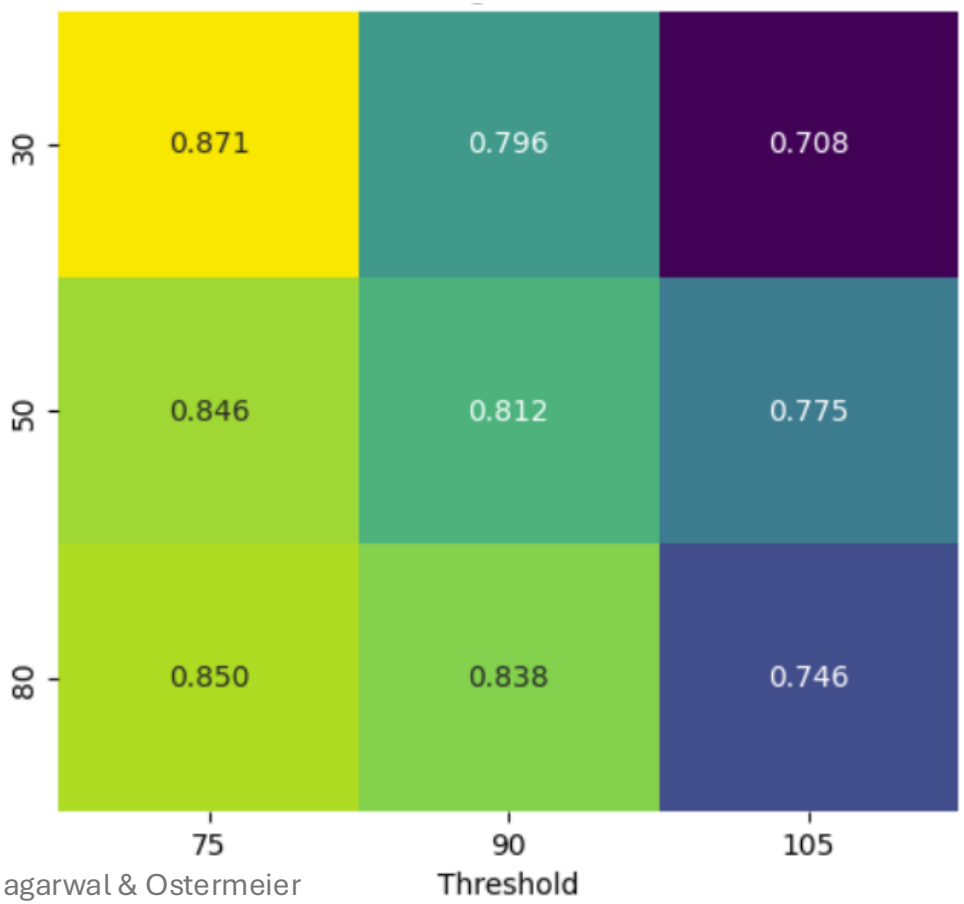


Choosing binary threshold and bin count

PLUS dataset

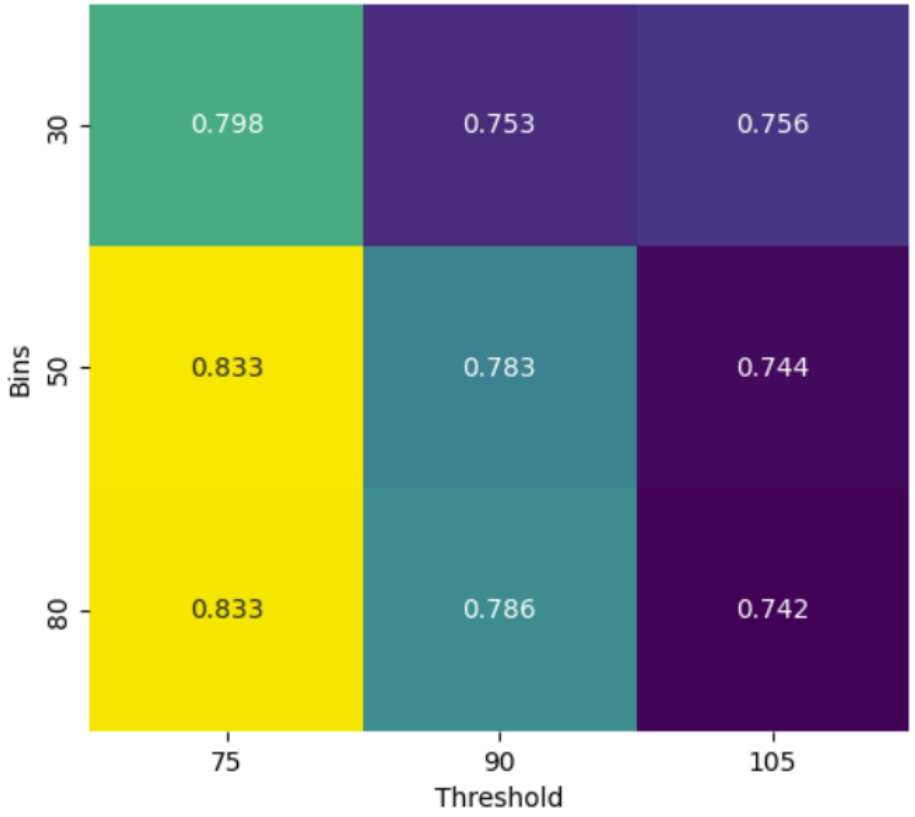


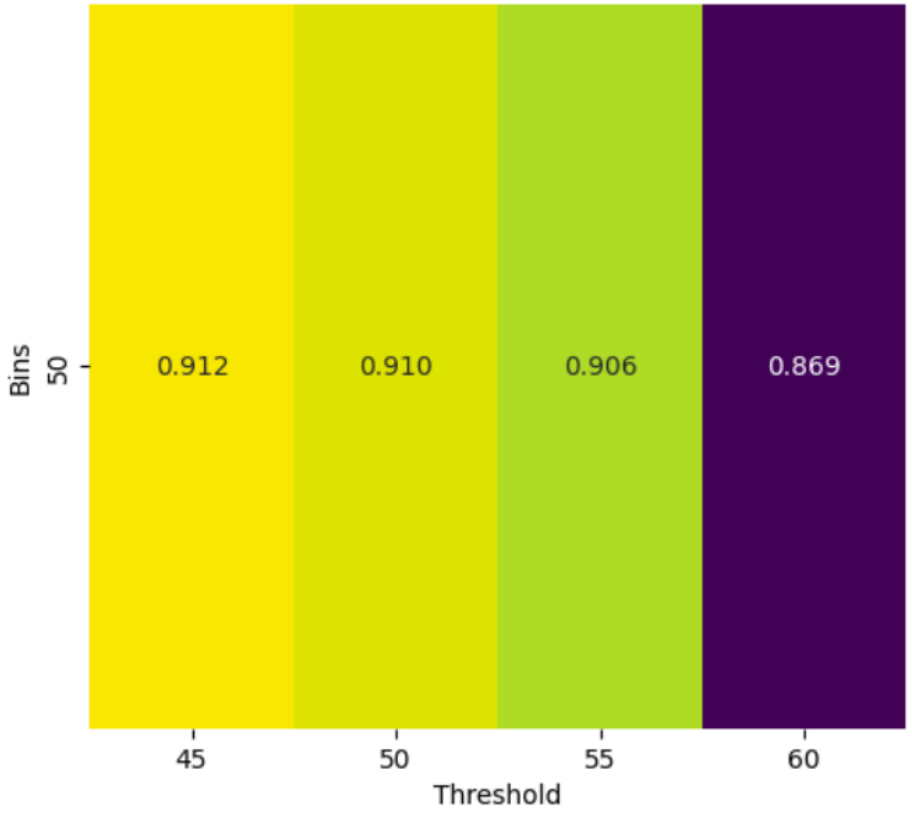
IDIAP dataset



Choosing binary threshold and bin count

SCUT dataset

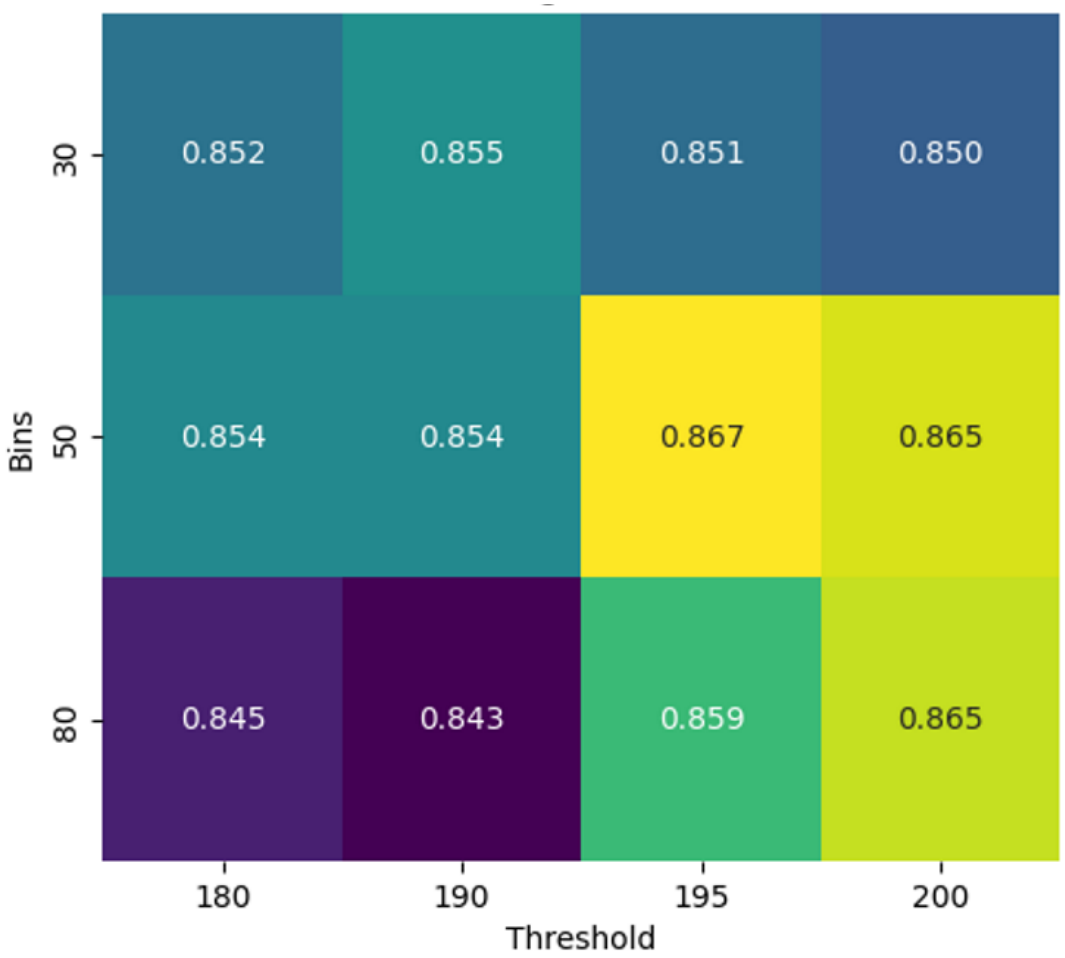




Threshold 75

Choosing binary threshold and bin count

SCUT dataset



Threshold 195

Results – Edge Region FD - PLUS

Binary threshold: 105
Bins: 50

Training set: Original	Accuracy	ACER	APCER	BPCER
Step 0 (4/5 training)	0.8742	0.126	0.053	0.198
Step 1 (2/5 training)	0.8547	0.145	0.06	0.23
Step 2 (1/5 training)	0.8398	0.16	0.071	0.249

Training set: VAE + original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.8636	0.136	0.038	0.235
Step 4 (3/5 synthetic + 1/5 orig)	0.8131	0.187	0.028	0.345
Step 5 (4/5 synthetic)	0.7961	0.204	0.031	0.377

Training set: AE + original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.8656	0.134	0.041	0.228
Step 4 (3/5 synthetic + 1/5 orig)	0.8254	0.175	0.036	0.313
Step 5 (4/5 synthetic)	0.8141	0.186	0.031	0.341

Results – Edge Region FD - SCUT

Binary threshold: 195
Bins: 50

Training set: Original	Accuracy	ACER	APCER	BPCER
Step 0 (4/5 training)	0.8729	0.127	0.185	0.069
Step 1 (2/5 training)	0.8507	0.149	0.206	0.093
Step 2 (1/5 training)	0.8285	0.172	0.246	0.097

Training set: VAE + original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.8382	0.162	0.132	0.192
Step 4 (3/5 synthetic + 1/5 orig)	0.7986	0.201	0.06	0.343
Step 5 (4/5 synthetic)	0.5000	0.5	0	1

Training set: AE + original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.8250	0.175	0.122	0.228
Step 4 (3/5 synthetic + 1/5 orig)	0.7674	0.233	0.058	0.407
Step 5 (4/5 synthetic)	0.5000	0.5	0	1

Results – Edge Region FD - IDIAP

Binary threshold: 70
Bins: 30

Training set: Original	Accuracy	ACER	APCER	BPCER
Step 0 (4/5 training)	0.8750	0.125	0.075	0.175
Step 1 (2/5 training)	0.8292	0.171	0.067	0.275
Step 2 (1/5 training)	0.8208	0.179	0.117	0.242

Training set: VAE + original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.8670	0.133	0.058	0.208
Step 4 (3/5 synthetic + 1/5 orig)	0.8542	0.146	0.025	0.267
Step 5 (4/5 synthetic)	0.8500	0.15	0.025	0.275

Training set: AE + original	Accuracy	ACER	APCER	BPCER
Step 3 (1/5 synthetic + 1/5 orig)	0.8750	0.125	0.025	0.225
Step 4 (3/5 synthetic + 1/5 orig)	0.8625	0.138	0.025	0.25
Step 5 (4/5 synthetic)	0.8541	0.146	0.025	0.267

Discussion

- As expected, reduced training increases errors
- Main source of errors: real misclassified as spoof (BPCER) → **usability issue**
  - **Segment-wise (Approach 1)**
    - PLUS: <1%
    - SCUT: <10%
    - IDIAP: up to ~35%
    - VAE performs slightly better for PLUS and IDIAP
  - **Edge region (Approach 2)**
    - PLUS: ~25% in reduced training, up to 40% with synthetics
    - IDIAP: 20-30%
    - SCUT: - Security issue: up to 25% APCER in reduced training
      - not usable with synthetic data: 100% misclassification of reals as spoof
    - AE performs slightly better for PLUS and IDIAP

Conclusion

Behavior with synthetic data:

- **Segment-wise (Approach 1)**
  - Slight increase in performance
  - Better distinguishable from real
  - For SCUT and PLUS data, spoofs could be replaced by synthetics without major damage
- **Edge region (Approach 2)**
  - Degradation of performance → more similarity in feature space of synthetics to real than spoof
  - Effect noticeable in all data sets with varying intensity, very dramatic in SCUT

BPCER
0.192
0.343
1

BPCER
0.228
0.407
1

References

- https://www.cosy.sbg.ac.at/~uhl/synthPAD_WIFS.pdf
- https://github.com/pranurs/fractal-dimensions

Thanks for your attention!